

Fault finder

Revise 360 lesson plan — 60 minutes

Unit	OCR J277 2.1 Algorithms — Paper 2, Computational thinking, algorithms and programming
Lesson	Fault finder (lesson 10 of the unit)
Lesson ID	al-l10
Experience ID	al-l10
Plan ID	PLAN-AL-L10
Content version	1f2b990f8974
Website	https://revise360.co.uk/experience.html?id=al-l10

Big question, outcomes and success criteria

Big question — What kinds of error occur when programming, and how are they found?

Prerequisites

The lesson before this one is al-l09. Pupils should already have finished it: the starter below retrieves from it.

Earlier in the unit: al-l01, al-l02, al-l03, al-l04, al-l05, al-l06, al-l07, al-l08.

Assessable outcomes

1. **al-l10-o1** — know what a syntax error is
2. **al-l10-o2** — know what a logic error is
3. **al-l10-o3** — be able to identify errors in code and suggest fixes

Vocabulary, resources and preparation

Key vocabulary — Syntax error, Logic error

Resources

SLIDES	presentations/AL_L10_FaultFinder_Lesson.pptx
WORKSHEET	worksheets/AL_L10_FaultFinder_Worksheet.docx
ONLINE EXPERIENCE	experience.html?id=al-l10

Before the lesson

1. Sign in on the **teacher dashboard** (teacher.html) and open **Class logins**. Enter the class or group and the usernames, one per line, then print the login cards. A pupil cannot sign in without one, so do this before the lesson rather than in it.
2. Print the worksheet, one per pupil.
3. Open the slides and the experience once yourself, on the machine you will be teaching from, so the 360 image is already cached.
4. Check headsets or devices are charged and on the school network.

How pupils get in

1. Go to `topics.html` on `revise360.co.uk`.
2. Sign in with the **username and PIN from the login card**. The class is already set against the card.
3. Choose the topic, then this lesson's card.

Worth knowing before you promise anything. In this build `APP_CONFIG.backendUrl` is empty, so logins, login cards and progress are stored on the device the pupil used. The teacher dashboard says so too. A pupil who moves to a different machine starts again, so keep a class on the same devices within a unit, and treat the paper worksheet as the record that travels.

Teaching sequence

Sixty minutes including transitions. The online block is the one to move if your lesson is shorter: see the adaptation at the end.

MIN	PHASE	TEACHER	PUPILS	EVIDENCE TO CHECK
0-5	Retrieve	Starter on the board or the worksheet, from the last lesson and this lesson's prerequisites.	Answer the starter on the worksheet.	The starter box on <code>AL_L10_FaultFinder_Worksheet.docx</code> is filled in.
5-8	Review the starter	Take answers; correct the one most got wrong.	Correct their own starter.	A correction written, not just a tick.
8-18	Explain and model	The new idea in two or three steps, with the example worked in front of the class.	Listen and answer questions about the example.	They can say why, not only what.
18-21	Set up and settle	Sign-in, launch, and what to write before anyone taps an answer.	Sign in, open the lesson, find Syntax errors.	Everyone is in and on station 1.
21-43	Online: work the stations in order	Pupils read each wall, write the key fact and the challenge, then answer at the badge.	Read each wall, write the key fact and the challenge, then answer at the badge.	The worksheet has writing on it for each of the 6 stations before the badge is tapped.
43-48	Check understanding	The hinge question, hands down.	Answer the hinge question, hands down.	The split across the options.
48-56	Apply	Exam practice on the worksheet, in silence.	Exam practice, in silence.	worksheet, Exam practice a) to c)
56-60	Exit and set up the next lesson	The exit question and the retrieval link to next lesson.	Write the exit answer.	One sentence that answers the big question.

The explanation and the worked example

This is what the walls of the experience say, station by station. Teach the first two or three from the front before anyone puts a headset on; the rest the pupils read for themselves.

1. Syntax errors

- A syntax error breaks the rules of the language.
- The program will not run at all: the translator reports the problem.
- Typical causes: a missing bracket, a missing quotation mark, a misspelt keyword.

Challenge on this wall: Write three examples of syntax errors in a language you use.

2. Logic errors

- A logic error follows the rules of the language but does the wrong thing.
- The program runs to the end and produces an answer, which is wrong.
- Typical causes: the wrong operator, the wrong order of tests, a loop that runs the wrong number of times.

Challenge on this wall: Describe a logic error that would not stop a program running.

3. Telling them apart

- Ask one question: does the program run?
- If it will not run, it is a syntax error.
- If it runs but the answer is wrong, it is a logic error.

Challenge on this wall: Sort six mistakes into syntax and logic errors.

4. Finding the line

- For a syntax error, the translator usually names the line, or the one just after it.
- For a logic error, work through the program by hand with a known input.
- Compare what should happen at each line with what actually happens.

Challenge on this wall: Work through a short program by hand and find where it goes wrong.

5. Suggesting a fix

- A fix should say what to change and what to change it to.
- 'Correct the error' earns nothing: 'change > to >=' earns the mark.
- Check the fix does not break another case.

Challenge on this wall: Write a fix for a loop that runs one time too many.

6. Answering error questions

- State the line number, the type of error, and the fix.
- Use the words syntax and logic: the marks are attached to them.
- If the question gives line numbers, always quote them.

Challenge on this wall: Plan a four-mark answer identifying and fixing an error.

Retrieval and the check for understanding

Starter — Write down the last time a program of yours didn't work. Did it refuse to run, or run and do the wrong thing?

Answer space: 3 lines on the worksheet.

The answer is in the teacher answers for this unit, not on this plan: a plan left on a desk should not give it away.

Hinge question

Use this one, hands down, before anyone moves on to independent work. It is a real activity from the experience (a1-l10-s4-c2), so the wording a pupil sees matches the wording you say.

Why work through a program with a specific input?

Correct: It shows exactly where what happens stops matching what should happen

What a pupil might choose instead

- It makes the program faster
- It fixes syntax errors
- It removes the need for testing

Count the hands on each. If more than a few of the class are on one wrong option, that option is the misconception to teach against before anyone goes on: go back to the station this question sits on and work through its example again, aloud, before the independent work.

The course's own response to a wrong answer here: That is what a trace table is for, in the next lesson.

One response, not four. one response covers every wrong option; it does not say what to revisit for each one, so decide your own reteach for each option before the lesson - the list above is the start of it.

Guided to independent practice

STAGE	ACTIVITIES	WHAT IT IS
Recognise	6	The answer is on the screen and the pupil picks it: multiple choice, choosing several, matching a pair.
Construct	2	Nothing is offered. The pupil classifies, orders, fills in a table or works a number out, and types or places the answer.

Scaffolds already in the product, and how they fade

- **Hint** is a ladder, not a door: the first rung names the idea, the next shows the shape of the code, the next how it starts, the last animates the technique on different data. A pupil asks for each rung. None of them is the answer.

- **Run** is free and unmarked; **Check my answer** marks. A pupil may run as often as they like, which is what makes trial and error safe.
- **I need help** says to ask you, and says plainly that the question still has to be completed. It does not notify anyone and it does not unlock anything.
- The worked example appears on every early activity and on none of the late ones. That fade is deliberate; it is not a missing example.

Accessibility

- **Read aloud** on every task window reads the task and its brief, with the prescribed values read as values rather than as punctuation.
- Text is 16px or larger throughout, and a question window scrolls rather than shrinking its type.
- Prescribed data is boxed and in a monospace face as well as coloured, so it still reads as code in black and white, in a high-contrast mode, and to a pupil who does not separate green from amber.
- **Lower literacy and EAL:** pre-teach the vocabulary above; let the pupil say the key fact before writing it; accept the key fact in two or three words.
- **SEND:** the sequence is compulsory but the pace is not. A pupil may finish a station next lesson; the progress saves after every answer and reopens at their earliest unfinished question.
- A headset is never required. Every experience works on a laptop or a tablet with a mouse or a finger.

Delivering the online experience

Before anyone enters

- Worksheet out, pen in hand. The rule is write first, then answer at the badge: the first answer is the one that counts.
- Say where the stations are. They are numbered around the room: 1 and 2 to the right, 3 and 4 behind, 5 and 6 to the left, and the final challenge on the floor.
- Say that a badge marked **Not yet** is not a fault. It opens when the station they are on is finished.

The stations, and what to record at each

#	STATION	ACTIVITIES	RECORD ON PAPER
1	Syntax errors	2	A syntax error is...
2	Logic errors	2	A logic error is...
3	Telling them apart	2	To tell the two apart, I ask...
4	Finding the line	2	To find an error, I...
5	Suggesting a fix	2	A good fix says...
6	Answering error questions	2	To answer an error question, I should...
final	Final challenge: debug it	2	the answer to the station's challenge

Where to pause

Stop the class after station 2 and take answers to its challenge out loud. That is the point where a pupil who has misread the technique can still be put right cheaply.

Afterwards

Headsets off and down. Ask three pupils for the key fact from a station of your choosing, in their own words and not read off the sheet. Then the exit question.

Assessment, evidence and unfinished work

ID	QUESTION	MARKS	WHERE
a1-l10-x1	State the difference between a syntax error and a logic error.	2	worksheet, Exam practice a); deck, Exam practice slide
a1-l10-x2	A program runs but always outputs one more than it should. What type of error is this? Explain your answer.	3	worksheet, Exam practice b); deck, Exam practice slide

ID	QUESTION	MARKS	WHERE
a1-l10-x3	Identify the error in the following program, state its type, and describe how to correct it.	3	worksheet, Exam practice c); deck, Exam practice slide

Expected answers

Model answers and marking guidance are teacher-only and are not printed on this plan. They are in the unit's teacher guide pack, and the per-activity marking for this lesson is in `answers/records/a1-l10.json`, which is not published with the website.

Common errors

The wrong options the activities offer are the errors this lesson expects. The commonest to watch for:

- **What happens when a program contains a syntax error?** — It runs but gives the wrong answer; It runs more slowly
- **Why are logic errors harder to find than syntax errors?** — They are always longer; They only happen in long programs
- **A program outputs 11 when it should output 10. What kind of error is it** — A syntax error; A translation error
- **Why work through a program with a specific input?** — It makes the program faster; It fixes syntax errors
- **Which is the strongest suggested fix?** — Fix line 2; Change the operator

Unfinished work

Record where each pupil stopped, not a tick. They resume at their earliest unfinished question, and a station left half-done is the thing to open next lesson. Do not mark the lesson complete for a pupil who did not finish it: the score screen already shows what was and was not done, and that is the honest record.

Exit, next lesson and homework

Exit check

One of these, written, not a show of hands:

- Syntax errors
- Logic errors
- Finding the line and suggesting a fix

The confidence columns on the worksheet are a supplement to this, not a substitute: a pupil who rates themselves confident and cannot answer the exit question has told you nothing.

Retrieval link to next lesson

Next is a1-l11, Trace room. Use this lesson's big question as next lesson's starter, so the first thing they do is retrieve it.

Homework

Finish any station left open, on the website, from the question they stopped at. If the class has no devices at home, set the exam practice on the worksheet instead and mark it next lesson.

Shorter lessons, longer lessons and no internet

A 40 to 45 minute lesson

Keep the starter, the explanation and the model at full length: cutting those is what makes the online work fail. Take the time out of the online block and the apply phase. Pupils stop where they are; their progress saves after every answer and reopens at their earliest unfinished question, so the next lesson starts there.

If the internet or the devices are unavailable

Teach from the slides and the worksheet. The slides carry the same teaching as the station walls, and the worksheet carries the key facts, the challenges and the exam practice. Set the online stations when the devices are back.

There is no offline version of the experience, and this plan does not contain one. Do not hand out a printed walkthrough of the stations as a substitute: the experiences are used on the website.